

Geometry-Only Plateau Selection for LPS

A support-size and chart-dimension selector for local PCA charts

Source:
~/current_projects/geosmooth/dev/methods/lps/status/
lps_geometry_only_plateau_kd_selector.tex

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1 Purpose

This note defines the geometry-only `plateau_kd` selector for local polynomial smoothing (LPS). The selector chooses a support size k and a local PCA chart dimension d before looking at the response y . Its purpose is to test whether a stable geometric dimension plateau can provide a useful deployable alternative to response-CV selection over the full Cartesian grid of (k, d) candidates.

The selector is deliberately conservative in its first implementation. It is not intended to replace `full_kd`; it is a new comparator. The important distinction is that `full_kd` asks which (k, d) has the best inner-CV prediction score, while `plateau_kd` asks which (k, d) is suggested by the input geometry alone.

2 The local chart objects

Let

$$X = \{x_1, \dots, x_n\} \subset \mathbb{R}^p$$

be the observed data cloud. For an anchor a , the support of size k is

$$U_a(k) = \{x_a\} \cup \{\text{the } k-1 \text{ nearest neighbors of } x_a\}.$$

The centered support matrix is

$$Z_a(k) = [(x_i - x_a)^\top]_{x_i \in U_a(k)}.$$

Let

$$\sigma_{a1}(k) \geq \sigma_{a2}(k) \geq \dots \geq 0$$

be the singular values of $Z_a(k)$. The total-variance-explained dimension at threshold τ , currently $\tau = 0.95$, is

$$d_a(k; \tau) = \min \left\{ d : \frac{\sum_{j=1}^d \sigma_{aj}(k)^2}{\sum_j \sigma_{aj}(k)^2} \geq \tau \right\}.$$

This dimension is clipped to the feasible design range. If the local polynomial degree is g , the number of polynomial columns is

$$q(d, g) = \binom{d+g}{g}.$$

With design margin m , a candidate is feasible only when

$$q(d, g) + m \leq k.$$

The current implementation also bounds d by the user-supplied numeric `chart.dim.grid`; for example, the CSD comparisons use $d \in \{1, \dots, 8\}$. This keeps `plateau.kd` inside the same numeric universe as `full.kd`.

3 The plateau rule

Let

$$\mathcal{K} = \{k_1 < k_2 < \dots < k_L\}$$

be the support-size grid. For each representative anchor a , compute the dimension path

$$d_a(k_1; \tau), d_a(k_2; \tau), \dots, d_a(k_L; \tau).$$

The baseline dimension is

$$d_a^0 = d_a(k_1; \tau).$$

The plateau endpoint is the last support size in the initial stable prefix:

$$k_a^* = \max \{k_\ell : d_a(k_r; \tau) = d_a^0 \text{ for every } r \leq \ell\}.$$

This is an initial-prefix rule. If the dimension changes and later returns to d_a^0 , the later return is ignored. This makes the rule closer to the simple geometric question: how far can the neighborhood grow before the local dimension estimate first changes?

For a global LPS fit, the anchor-specific quantities are aggregated by medians:

$$k^* = \text{nearest}_{k \in \mathcal{K}} \text{median}_a k_a^*, \quad d^* = \text{clip}(\text{round median}_a d_a^0).$$

The clipping enforces the supplied dimension grid, the ambient dimension, and the polynomial-design feasibility inequality at k^* . The resulting candidate family is therefore

$$\mathcal{G}_{\text{plateau}} = \{(k^*, d^*)\},$$

with one row for each requested degree, kernel, and bandwidth multiplier. In the CSD comparison reported with this note, the degree, kernel, and bandwidth multiplier are fixed, so $\mathcal{G}_{\text{plateau}}$ contains one candidate.

4 Comparison with existing selectors

Selector	Meaning
<code>auto</code>	Uses observed X to estimate one global chart dimension d , then selects support size k by the ordinary LPS CV grid. The chosen dimension is shared by all anchors.
<code>local.auto</code>	Uses observed X to estimate a dimension d_a separately for each anchor. The support size is still selected by the ordinary LPS CV grid.
<code>full.kd</code>	Evaluates the full Cartesian numeric grid $\mathcal{G}_{\text{full}} = \mathcal{K} \times \mathcal{D}$ by inner CV and chooses the one global pair (k^*, d^*) with the best CV score.
<code>sparse.kd</code>	Evaluates a smaller deterministic skeleton of the same Cartesian grid by inner CV. It is a response-CV shortcut to <code>full.kd</code> .
<code>plateau.kd</code>	Uses X only to choose one global (k^*, d^*) from the initial dimension plateau. CV is still reported for the selected candidate, but CV does not choose the pair.

5 What the comparison should reveal

The `plateau_kd` selector tests a specific hypothesis. If the dominant mistake in `full_kd` is noisy inner-CV ranking over many nearby candidates, then a geometry-only plateau may sometimes perform comparably while being much cheaper and more stable. If the response really needs a support scale different from the geometry plateau, then `plateau_kd` should lose to `full_kd`, and the loss should be visible in truth-facing RMSE.

The associated HTML comparison therefore reports, for each synthetic dataset,

$$R_m = \left\{ \frac{1}{n} \sum_{i=1}^n \left(\hat{f}_m(x_i) - f_i \right)^2 \right\}^{1/2},$$

where f_i is the known synthetic truth and $\hat{f}_m$ is the fitted LPS estimate under method m . The report also records the selected support size, selected chart dimension, observed CV score, and runtime. In the 1D and 2D examples the truth and fitted functions are plotted directly; in higher dimensions the report uses summary tables because direct surface display is not meaningful.

6 Limitations

The current `plateau_kd` implementation is intentionally a first, auditable rule. It uses the total-variance-explained dimension only; it does not yet combine variance thresholds with eigengaps, participation rank, or a response-facing near-tie rule. It aggregates local plateau endpoints into one global (k, d) , so it is not an anchor-specific local-scale method. A later local variant could use k_a^* and d_a^0 anchor by anchor, but that would be a different selector and should be tested separately.

7 Implementation status

The selector is implemented through the shared coupled candidate-specification path. The public trigger is

```
selection.strategy = "plateau_kd"
```

with `coordinate.method = "local.pca"` and an explicit numeric `chart.dim.grid`. The fit object records the strategy, the selected candidate, and per-anchor plateau diagnostics under

```
fit$diagnostics$coupled.kd.selection.
```

The implementation includes a unit test that verifies the one-candidate LPS path and verifies that the selector refuses to run without an explicit numeric dimension universe.